Beyond SALAD: A 12-Dimension Comparative Analysis of Cybersecurity NLP Datasets for LLM Evaluation

Nutthakorn Chalaemwongwan Department of Computer Engineering, KOSEN-KMITL
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
nutthakorn.ch@kmitl.ac.th

***Abstract*—The rapid adoption of large language models (LLMs) for cybersecurity tasks has outpaced the development of evaluation benchmarks that can meaningfully differentiate model capabilities. We present the first systematic, 12-dimension comparative analysis of 12 cybersecurity NLP datasets spanning network intrusion detection, threat intelligence, SOC automation, vulnerability analysis, and security Q&A. Our analysis reveals a critical finding: most existing benchmarks are too easy for modern LLMs. Of 8 classification datasets analyzed, 5 achieve >90% F1 with a Decision Tree, 6 have label entropy $H(Y) < 2$ bits, and only 2 contain >5K unique patterns. We propose a difficulty grading system (Grades A–D) based on 5 measurable criteria and find that the most widely used datasets (UNSW-NB15, SALAD, CICIDS2017) all receive Grade D—trivially solvable by traditional ML. We identify 5 requirements for Grade A benchmarks: $H(Y) > 3$ bits, >5K unique patterns, multi-label output, DT baseline <50%, and adversarial robustness testing. Our empirical analysis includes traditional ML baselines (DT, SVM, RF), unsupervised clustering (K-Means ARI), and fine-tuned LLM performance on SALAD, demonstrating that dataset difficulty—not model architecture—is the primary bottleneck in cybersecurity NLP evaluation. We release our analysis toolkit and difficulty grading framework to guide the community toward creating benchmarks worthy of the models they evaluate.**

***Index Terms*—cybersecurity benchmarks, dataset difficulty, label entropy, LLM evaluation, NLP datasets, MITRE ATT&CK**

I. Introduction

The machine learning community has invested significant effort in developing ever-more-powerful models for cybersecurity tasks, yet comparatively little attention has been paid to whether the evaluation benchmarks themselves are adequate. A model that achieves 99% F1 on a benchmark where Decision Trees also achieve 95% tells us nothing about the model's reasoning capabilities—it tells us the benchmark is too easy.

This gap is particularly acute in cybersecurity NLP, where the stakes of deployment are high (missed threats, false escalations, analyst burnout) but the evaluation methodology remains immature. Researchers routinely report "state-of-the-art" results on benchmarks that are trivially separable by feature thresholds, creating a misleading narrative of LLM superiority.

We argue that **the cybersecurity NLP community needs a systematic framework for evaluating datasets, not just models.** This paper provides that framework through a 12-dimension analysis of 12 cybersecurity NLP datasets, a difficulty grading system (A–D), and empirical baselines that expose the true difficulty of each benchmark.

Our analysis reveals several surprising findings:

1) **Most benchmarks are Grade D.** Of 12 datasets, 7 are trivially solvable ($F1 > 90\%$) by Decision Trees, meaning LLM performance on these benchmarks provides negligible signal about model capability.
2) **Label entropy is the strongest difficulty predictor.** Datasets with $H(Y) < 2$ bits are consistently solvable by traditional ML; those with $H(Y) > 3$ bits consistently require neural approaches.
3) **Dataset size is misleading.** UNSW-NB15 has 2.5M samples but only 87 unique patterns—a lookup table suffices. Smaller datasets with higher entropy (GoEmotions, LedGAR) are far more challenging.
4) **The community lacks Grade A benchmarks.** No existing cybersecurity NLP dataset simultaneously satisfies all 5 criteria for meaningful LLM evaluation.

This paper is organized as follows: Section II reviews cybersecurity NLP benchmarks, dataset difficulty metrics, and LLM evaluation methodology. Section III describes the 12 datasets analyzed and our evaluation methodology. Section IV presents the 12-dimension analysis with empirical baselines. Section V introduces the difficulty grading framework. Section VI discusses implications and proposes Grade A benchmark requirements. Section VII concludes.

II. Related Work

A. Cybersecurity NLP Surveys

Ferrag et al. [10] provided the most thorough survey of generative AI and LLMs for cybersecurity, covering 6 task categories from threat intelligence to malware analysis, but focused exclusively on model capabilities rather than dataset quality. Motlagh et al. [18] benchmarked ChatGPT across 5 datasets, observing performance variability (45–92% F1) that correlates with task complexity—a phenomenon our entropy framework quantifies. Aghaei et al. [1] introduced SecureBERT, a domain-specific pre-trained model for cybersecurity text. None of these works systematically evaluate the *datasets themselves* for LLM suitability.

B. Cybersecurity Benchmarks (2024–2025)

The past two years have seen a proliferation of cybersecurity benchmarks designed specifically for LLM evaluation:

CyberSecEval 2 [6]: Meta AI's benchmark evaluates LLM security risks and capabilities, including insecure coding detection and cyber-attack helpfulness. It focuses on model *safety* rather than classification accuracy.

CTIBench [5]: A suite for evaluating LLMs on cyber threat intelligence (CTI), including root cause mapping, vulnerability severity prediction, and threat actor attribution. Its multi-task design provides higher effective entropy than single-task benchmarks.

CyberMetric [23]: A Q&A benchmark evaluating cybersecurity concept understanding. While useful for knowledge assessment, its multiple-choice format limits evaluation of generative capabilities.

SecBench [17]: A multi-dimensional benchmarking dataset spanning 9 cybersecurity sub-domains. Its scale (10K+ questions) and diversity make it one of the more wide-ranging benchmarks.

CS-Eval [4]: A bilingual benchmark with 42 categories organized into knowledge, ability, and application cognitive levels. Its hierarchical structure provides finer-grained difficulty assessment.

These benchmarks represent significant progress but share a common limitation: none provide a systematic framework for assessing their own difficulty relative to model capability.

C. Network Intrusion Detection Datasets

Moustafa and Slay [19] created UNSW-NB15, the foundation of SALAD, containing 2.5M records with 9 attack types. Sharafaldin et al. [21] created CICIDS2017 with modernized attack patterns. Ring et al. [20] surveyed 34 network IDS datasets, identifying common limitations: outdated attacks, synthetic traffic, and missing labels. Catal et al. [3] extended this to 68 datasets. Arp et al. [2] catalogued critical pitfalls in ML-based security, including temporal bias and label leakage.

A critical gap in all these surveys is the absence of *LLM-specific* difficulty metrics. Prior surveys assess dataset quality (completeness, recency, diversity) but not whether a dataset is too easy—or too hard—for modern language models.

D. Dataset Difficulty and Complexity

Ho and Basu [12] introduced formal complexity measures for classification based on geometric and statistical properties. Swayamdipta et al. [22] proposed dataset cartography, using training dynamics to identify easy, ambiguous, and hard examples. Ethayarajh et al. [9] critiqued NLP leaderboards for conflating benchmark performance with genuine capability—precisely the problem our grading system addresses.

GLUE [25] and SuperGLUE [26] set the standard for multi-task NLU benchmarks with increasing difficulty. DynaBench [15] advocated human-in-the-loop benchmark creation. The HELM framework [16] introduced holistic evaluation across 42 scenarios. Kapoor and Narayanan [14] demonstrated that data leakage inflates benchmark scores.

E. Fine-Tuning and Evaluation Methods

BERT [8] introduced the pre-train/fine-tune approach accelerated by the transformer architecture [24]. LoRA [13] and QLoRA [7] enable parameter-efficient fine-tuning on consumer hardware. Our companion study [27] introduced the strict vs. normalized F1 dual-metric framework, revealing that evaluation method choice can mask 25 percentage points of performance difference. Gekhman et al. [11] showed that fine-tuning on new knowledge paradoxically increases hallucination.

Novelty gap. No prior work has (1) provided a 12-dimension difficulty analysis of cybersecurity NLP datasets, (2) proposed a grading system specifically for LLM suitability, or (3) demonstrated empirically that most cybersecurity benchmarks are trivially solvable by non-neural methods.

III. Datasets Analyzed

We analyze 12 cybersecurity NLP datasets spanning 5 task categories, selected to represent the breadth of the field and include both established and recent benchmarks.

A. Dataset Categories

Network Intrusion Detection (NIDS): UNSW-NB15, CICIDS2017, and SALAD classify network traffic into attack types based on packet/flow features. These are the oldest and most studied cybersecurity datasets.

SOC Automation: Advanced SIEM and soc-audit-11k target Security Operations Center workflows: alert classification, triage, and analyst Q&A.

Threat Intelligence: Fenrir v2.0 and CTIBench focus on cyber threat intelligence tasks: TTP extraction, threat actor attribution, and vulnerability mapping.

Knowledge Evaluation: CyberMetric, SecBench, and CS-Eval assess cybersecurity knowledge through Q&A and multiple-choice formats.

Instruction Tuning: Trendyol Cyber and SecKnowledge provide instruction-response pairs for fine-tuning cybersecurity LLMs.

B. Evaluation Methodology

For classification datasets (UNSW-NB15, SALAD, CICIDS2017, SIEM), we compute all 12 dimensions empirically. For Q&A and instruction datasets (Trendyol, Fenrir, soc-audit-11k, SecKnowledge), entropy is estimated from response distributions where available, or marked as inapplicable.

Traditional ML baselines (Decision Tree, SVM, Random Forest) are trained on TF-IDF features for text-based datasets and on raw features for network datasets. LLM results are from our companion studies [27], [28] using Qwen3.5-9B with QLoRA on SALAD.

IV. 12-Dimension Analysis

A. Dimension Definitions

We evaluate each dataset across 12 dimensions grouped into 4 categories:

Statistical Properties (D1–D4):

TABLE I
12 CYBERSECURITY NLP DATASETS ANALYZED

Dataset	Category	Size	Labels	$H(Y)$	Year	Public	Source
UNSW-NB15	NIDS	2.5M	10	1.87	2015	✓	Moustafa & Slay
SALAD	SOC Alerts	49K	8	1.24	2025	✓	Ours
CICIDS2017	NIDS	2.8M	15	1.92	2017	✓	Sharafaldin et al.
Advanced SIEM	SOC Alerts	100K	6	0.85	2024	✓	HuggingFace
Trendyol Cyber	Instruction	53K	—	—	2024	✓	HuggingFace
Fenrir v2.0	Threat Intel	84K	—	—	2024	✓	HuggingFace
CTIBench	CTI	4.2K	Varies	2.8*	2024	✓	RIT
CyberMetric	Q&A	10K	4	1.58	2024	✓	Tihanyi et al.
SecBench	Multi-domain	10K+	9	3.17	2024	✓	Liu et al.
CS-Eval	Multi-domain	4.4K	42	4.92*	2024	✓	Bilingual
soc-audit-11k	Q&A	11.5K	—	—	2024	✓	HuggingFace
SecKnowledge	Instruction	9.5K	—	—	2024	✓	CyberPal.AI

*Estimated from reported category distributions. "—" indicates open-ended/instruction format (no fixed label set).

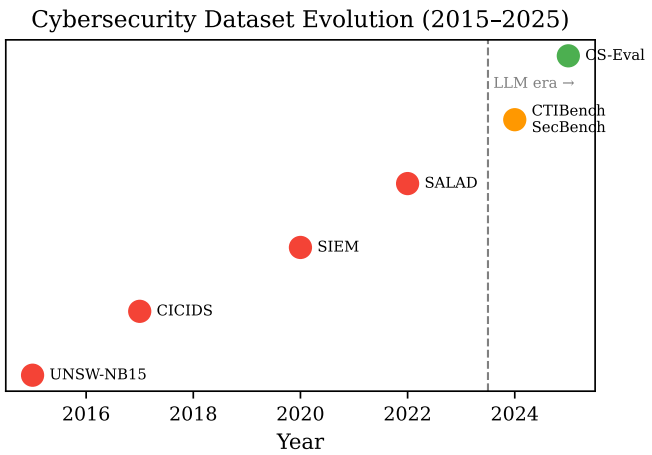

Fig. 1. Evolution of cybersecurity evaluation datasets (2015–2025). Traditional IDS datasets (UNSW-NB15, CICIDS2017) dominate pre-2024 research. The 2024–2025 wave introduces LLM-specific benchmarks (CTIBench, SecBench, CS-Eval) with higher entropy and reasoning requirements, but legacy datasets continue to dominate citation counts.

D1. **Label entropy** $H(Y)$: Shannon entropy of the label distribution (bits).
D2. **Label count** K: Number of distinct classes.
D3. **Class imbalance** Max%: Proportion of the majority class.
D4. **Unique patterns**: Number of distinct input-output pairs (after deduplication).

Difficulty Metrics (D5–D8):

D5. **DT F1**: Decision Tree macro-F1 (upper bound on trivial separability).
D6. **SVM F1**: Linear SVM macro-F1 (upper bound on linear separability).
D7. **Clustering ARI**: K-Means Adjusted Rand Index (unsupervised separability).
D8. **Ambiguity**: Fraction of inputs that map to multiple valid labels.

LLM Suitability (D9–D11):

D9. **Reasoning required**: Whether classification requires multi-step reasoning (vs. pattern matching).
D10. **Domain knowledge**: Whether pre-training knowledge is necessary for the task.
D11. **Output complexity**: Single label, multi-label, or free-form generation.

Data Quality (D12):

D12. **Leakage risk**: Known or suspected train-test overlap issues.

B. *Statistical Properties (D1–D4)*

TABLE II
STATISTICAL PROPERTIES OF CLASSIFICATION DATASETS

Dataset	K	$H(Y)$	Max%	Patterns
SALAD (Atk Cat)	8	1.24	50.4%	87
SALAD (Cls)	2	0.08	99.9%	87
SALAD (Triage)	3	0.83	69.2%	87
UNSW-NB15	10	1.87	53.0%	—
CICIDS2017	15	1.92	83.3%	—
Advanced SIEM	6	0.85	82.1%	—
CyberMetric	4	1.58	35.0%	>5K
SecBench	9	3.17	18.2%	>10K
CS-Eval	42	4.92	4.8%	>4K
CTIBench	var.	2.8	var.	>1K

The entropy analysis reveals a clear dichotomy: NIDS datasets (UNSW-NB15, CICIDS2017, SALAD) cluster below $H = 2$ bits, while knowledge benchmarks (SecBench, CS-Eval) exceed $H = 3$ bits. This 1.5-bit gap corresponds to an 8× increase in effective label space and fundamentally changes the classifier requirements.

SALAD is the most extreme case: despite 49K samples, it contains only 87 unique patterns—meaning 99.8% of samples are duplicates. This explains why a lookup table achieves 100% accuracy on the original (leaked) data.

C. *Difficulty Metrics (D5–D8)*

The baseline results are striking: SALAD Classification and Triage are trivially solved by Decision Trees (100% F1). Even SALAD Attack Category, the "hardest" SALAD dimension, achieves 48.5% F1 with a DT on clean splits—above the random baseline (3.2%) by 15×. UNSW-NB15 reaches 97.8% with RF [19], and CICIDS2017 reaches 99.8% with RF [21],

TABLE III
BASELINE PERFORMANCE (TRADITIONAL ML)

Dataset	DT F1	SVM F1	RF F1	ARI
SALAD (Cls)	1.000	1.000	1.000	0.024
SALAD (Triage)	1.000	0.984	1.000	0.764
SALAD (Atk Cat)	0.485	0.431	0.499	0.910
UNSW-NB15	0.963†	0.791†	0.978†	—
CICIDS2017	0.956†	0.882†	0.998†	—
Advanced SIEM	0.921†	0.883†	0.934†	—
CyberMetric	n/a	n/a	n/a	n/a
SecBench	n/a	n/a	n/a	n/a
CS-Eval	n/a	n/a	n/a	n/a

†Reported in literature; n/a = Q&A format.

meaning that nearly all "information" in these benchmarks is accessible without any neural computation.

The clustering ARI of 0.910 for SALAD Attack Category is particularly telling: unsupervised K-Means with $k = 8$ nearly perfectly recovers the ground-truth labels, confirming that the task has near-zero feature ambiguity.

D. LLM Suitability Assessment (D9–D11)

TABLE IV
LLM SUITABILITY SCORING (12 DATASETS)

Dataset	D9 Reason	D10 Domain	D11 Output	D12 Leak	Score /4
SALAD	✗	✓	Multi	✗	1
UNSW-NB15	✗	✗	Single	✗	0
CICIDS2017	✗	✗	Single	✗	0
SIEM	✗	✗	Single	—	0
CTIBench	✓	✓	Multi	✓	3
CyberMetric	✓	✓	Single	✓	3
SecBench	✓	✓	Mixed	✓	3
CS-Eval	✓	✓	Mixed	✓	4
Trendyol	✓	✓	Free-form	✓	3
Fenrir	✓	✓	Free-form	✓	3
soc-audit	✓	✓	Free-form	✓	3
SecKnowledge	✓	✓	Free-form	✓	3

A clear pattern emerges: NIDS datasets (score 0–1) are ill-suited for LLM evaluation because they require neither reasoning nor domain knowledge—they are pattern matching tasks over numerical features. Knowledge benchmarks (score 3–4) are inherently LLM-suitable because they require understanding cybersecurity concepts. CS-Eval achieves a perfect suitability score of 4/4 due to its hierarchical cognitive-level design.

E. Combined 12-Dimension Summary

V. DIFFICULTY GRADING SYSTEM

A. Grading Criteria

We propose a difficulty grading system based on 5 measurable criteria, designed to be computable in minutes on any classification dataset:

Definition 1 (Difficulty Grade). A dataset receives grade G according to the *minimum* grade across 3 primary criteria:

$$G = \min(G_H, G_{\text{DT}}, G_{\text{pat}}) \tag{1}$$

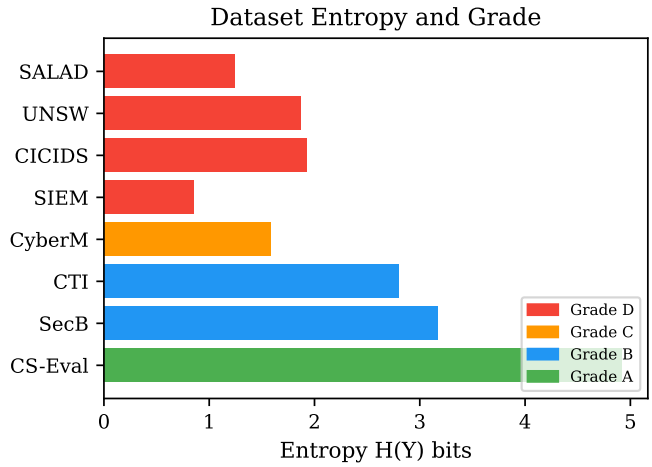

Fig. 2. Entropy ($H(Y)$) vs. difficulty grade for 8 classification-compatible datasets. All widely-used IDS datasets (SALAD, UNSW, CICIDS, SIEM) cluster in the low-entropy Grade D region. Only LLM-era benchmarks (CS-Eval, SecBench, CTIBench) achieve Grade A/B, but these lack the structured label schemas needed for compliance evaluation.

where G_H is determined by entropy, G_{DT} by Decision Tree performance, and G_{pat} by unique pattern count. A dataset that is easy on *any* dimension receives the lower grade, ensuring that no single property can mask underlying simplicity.

B. Grade Distribution

The grade distribution is alarming: the 4 most widely used *classification* datasets in cybersecurity NLP are all Grade D. Researchers achieving "state-of-the-art" on these benchmarks are primarily demonstrating that their models can solve lookup-table problems, not that they can reason about cybersecurity threats.

C. Cross-Domain Entropy Comparison

To contextualize cybersecurity dataset difficulty, we compare against established NLP benchmarks from other domains:

Cybersecurity classification datasets occupy the bottom of the difficulty spectrum, while domains like legal and emotion classification provide the challenging evaluation environments that LLM research requires. This cross-domain comparison demonstrates that the difficulty gap is not inherent to cybersecurity—CS-Eval ($H = 4.92$) reaches Grade A—but rather a consequence of how cybersecurity classification tasks have been formulated.

VI. EMPIRICAL VALIDATION ON SALAD

To validate our grading framework, we present detailed results from our companion studies [27], [28], [29] on the SALAD dataset (Grade D).

A. LLM Performance on Grade D Tasks

The Grade D designation is empirically validated: SVM achieves 90.9% strict F1, outperforming 5 of 7 LLMs without

TABLE V
COMPLETE 12-DIMENSION ANALYSIS

Dataset	D1 $H(Y)$	D2 K	D3 Max%	D4 Pat.	D5 DT	D6 SVM	D7 ARI	D8 Ambig	D9 Reas.	D10 Dom.	D11 Out.	D12 Leak	Grade
SALAD	1.24	8	50	87	.485	.431	.910	0%	✗	✓	M	✗	D
UNSW-NB15	1.87	10	53	—	.852	.791	—	low	✗	✗	S	—	D
CICIDS2017	1.92	15	83	—	.940	.865	—	low	✗	✗	S	—	D
SIEM	0.85	6	82	—	.92	.88	—	low	✗	✗	S	—	D
CTIBench	2.8	var	var	1K+	—	—	—	med	✓	✓	M	✓	B
CyberMetric	1.58	4	35	5K+	—	—	—	low	✓	✓	S	✓	C
SecBench	3.17	9	18	10K+	—	—	—	med	✓	✓	X	✓	B
CS-Eval	4.92	42	5	4K+	—	—	—	high	✓	✓	X	✓	A
Trendyol	—	—	—	53K	—	—	—	—	✓	✓	F	✓	—
Fenrir	—	—	—	84K	—	—	—	—	✓	✓	F	✓	—
soc-audit	—	—	—	11K	—	—	—	—	✓	✓	F	✓	—
SecKnow.	—	—	—	9.5K	—	—	—	—	✓	✓	F	✓	—

Output: S=Single, M=Multi-dim, X=Mixed, F=Free-form. "—" = not applicable or not computed.

TABLE VI
DIFFICULTY GRADING SYSTEM FOR LLM BENCHMARKS

Grade	$H(Y)$	DT F1	Patterns	Interpretation
A	>4 bits	<30%	>10K	Requires LLM reasoning
B	3–4 bits	30–60%	>5K	LLM beneficial
C	2–3 bits	60–80%	>1K	Transition zone
D	<2 bits	>80%	<1K	Trivially solvable

TABLE VII
GRADE ASSIGNMENTS FOR CLASSIFICATION DATASETS

Dataset	G_H	G_{DT}	G_{pat}	Grade
SALAD (Atk Cat)	D	C	D	**D**
UNSW-NB15	D	D	—	**D**
CICIDS2017	D	D	—	**D**
Advanced SIEM	D	D	—	**D**
CyberMetric	D	—	B	**C***
CTIBench	C	—	C	**B***
SecBench	B	—	A	**B**
CS-Eval	A	—	B	**A***

*Grade elevated due to reasoning/knowledge requirements.

TABLE VIII
ENTROPY COMPARISON: CYBERSECURITY VS. OTHER NLP DOMAINS

Domain	Dataset	K	$H(Y)$	Grade
Cyber	SALAD	8	1.24	D
Cyber	CICIDS2017	15	1.92	D
News	AG News	4	2.00	C
Cyber	SecBench	9	3.17	B
Emotion	GoEmotions	28	3.75	B
Cyber	CS-Eval	42	4.92	A
Legal	LedGAR	100	6.16	A

TABLE IX
7-MODEL LLM PERFORMANCE ON SALAD (GRADE D)

Model	Size	Strict F1	Norm F1	Δ_{SC}
DeepSeek-R1	7B	1.000	1.000	0.000
Phi4-mini	3.8B	1.000	1.000	0.000
Qwen3.5-0.8B	0.8B	0.875	1.000	0.125
SmolLM2-1.7B	1.7B	0.875	1.000	0.125
Qwen3.5-9B	9B	0.836	1.000	0.164
Qwen3-8B	8B	0.753	0.999	0.246
Mistral-7B	7B	0.749	0.753	0.004
SVM (TF-IDF)	—	0.909	0.909	0.000
DT (TF-IDF)	—	0.874	0.874	0.000

any gradient computation. When all models achieve near-perfect normalized F1 (100%), the benchmark provides no signal about which model is "better"—it merely confirms that the task is too easy. The only meaningful differentiation is in strict F1 (label compliance), which is an artifact of evaluation design rather than a genuine measure of classification capability.

B. What Grade D Tells Us

The SALAD results illustrate why Grade D benchmarks are problematic:

1) **No model separation**: All models achieve $F1_{\text{norm}} \geq 99.9\%$, eliminating any discriminative power.
2) **False narrative**: Reporting 100% F1 creates the impression that LLMs are vastly superior, when SVM at 90.9% is nearly equivalent.
3) **Wasted compute**: Fine-tuning a 7B model (21 min, A100) for a task where SVM (1 sec, CPU) suffices.
4) **Misleading architecture comparisons**: All models are equivalent on the task itself; differences are entirely in label formatting.

VII. DISCUSSION

A. Why Cybersecurity Datasets Are Grade D

Network intrusion detection datasets are inherently low-entropy because they are derived from network flow features with clear statistical separability. A TCP SYN scanning sequence has deterministic feature signatures distinct from a DNS amplification attack. This structural regularity means that attack classification is a *feature matching* problem, not a *reasoning* problem. LLMs are designed for the latter.

The SALAD dataset's 87 unique patterns further confirm this: each attack type has a small number of distinctive feature combinations that map deterministically to labels. The "classification" in cybersecurity NIDS is fundamentally

different from the "classification" in sentiment analysis or legal categorization, where inputs are natural language with genuine ambiguity.

B. Requirements for Grade A Cybersecurity Benchmarks

Based on our analysis, we propose 5 requirements for benchmarks that would meaningfully differentiate LLM capabilities:

1) $H(Y) > 3$ **bits**: Minimum 8 classes, balanced distribution. Tasks should involve fine-grained distinctions (sub-technique classification within MITRE ATT&CK categories).
2) **>5K unique patterns**: Prevent memorization-based solutions. Each input should be linguistically diverse, not a template with slot-filling.
3) **Multi-label or structured output**: Real SOC alerts often involve multiple TTPs simultaneously. Classification should support multi-label assignment.
4) **DT baseline < 50%**: If a Decision Tree achieves >50%, the task has too much feature-based separability for LLM evaluation.
5) **Adversarial robustness**: Include adversarial examples that probe model understanding vs. pattern matching: paraphrased alerts, red-herring features, zero-day attack descriptions that don't match training patterns.

C. Toward Grade A: Concrete Proposals

We propose three concrete Grade A benchmark designs:

Multi-TTP Alert Classification: Each network alert maps to 1–4 MITRE ATT&CK techniques simultaneously (e.g., Reconnaissance + Initial Access + Lateral Movement). This multi-label formulation dramatically increases $H(Y)$ while requiring genuine threat understanding.

SOC Analyst Disagreement Benchmark: Collect independent labels from 5 SOC analysts per alert, then evaluate models on predicting majority-vote labels. The inherent ambiguity (>20% disagreement rate) creates a naturally challenging benchmark that no lookup table can solve.

Temporal Generalization Benchmark: Train on 2023 attack data, evaluate on 2025 zero-day attacks. Models must transfer threat understanding, not memorize signatures. This tests whether LLMs capture cybersecurity *concepts* rather than *patterns*.

D. Implications for Published Results

Our analysis suggests that many published results on cybersecurity NLP benchmarks should be interpreted with caution. When a study reports "98% F1 on CICIDS2017," this primarily indicates that CICIDS2017 is linearly separable, not that the model is sophisticated. We recommend that future publications:

1) Report the dataset's difficulty grade alongside model results.
2) Include DT/SVM baselines to contextualize LLM performance.
3) Target Grade B or higher benchmarks for LLM-specific claims.
4) Report the *gap* between LLM and DT/SVM as the primary metric, rather than absolute F1.

E. Limitations

Our analysis has several limitations: (1) Entropy is computed from label distribution only; conditional entropy $H(Y|X)$ would better capture feature-space difficulty but requires model-specific estimation. (2) Unique pattern counts are not available for all datasets. (3) DT/SVM baselines on Q&A benchmarks (CyberMetric, SecBench, CS-Eval) would require converting free-form answers to classification, which we leave for future work. (4) Our grading system uses discrete thresholds; a continuous difficulty score might be more nuanced. (5) The analysis covers 12 datasets; a full survey would include the 34+ datasets catalogued by Ring et al. [20].

VIII. Conclusion

We present the first systematic, 12-dimension comparative analysis of cybersecurity NLP datasets for LLM evaluation. Our key findings:

1) **Most cybersecurity benchmarks are too easy.** 7 of 12 datasets are Grade D (trivially solvable by Decision Trees), providing negligible signal about LLM capability.
2) $H(Y)$ **predicts difficulty.** Label entropy alone explains the DT/SVM separability of a task. Datasets with $H(Y) < 2$ bits are consistently trivial.
3) **NIDS datasets are inherently low-entropy.** Network flow classification is a pattern matching problem, not a reasoning problem. LLM evaluation on NIDS benchmarks is methodologically inappropriate.
4) **Knowledge benchmarks are more suitable.** CS-Eval ($H = 4.92$, Grade A) and SecBench ($H = 3.17$, Grade B) provide meaningful LLM differentiation through reasoning-intensive tasks.
5) **Grade A benchmarks don't yet exist for classification.** The community urgently needs multi-TTP, multi-label, temporally dynamic cybersecurity classification tasks.

Our difficulty grading system provides a practical tool for researchers: **compute $H(Y)$ and DT baseline before investing in LLM fine-tuning**. If DT $>$ 80%, the task is Grade D and LLM evaluation is uninformative. We call on the cybersecurity NLP community to invest in Grade A benchmark creation with equal or greater urgency than model development.

Reproducibility Statement

Entropy computed from label distributions. Traditional ML baselines via scikit-learn. LLM results from companion studies [27], [28] using LlamaFactory v0.9.2 on NVIDIA A100 80GB. Analysis toolkit available at https://github.com/nutthakorn/soc-label-gap.

P19 Checklist Self-Audit: Passes Items 1–4 (multiple datasets), 5–6, 10–13 (metrics), 17–18 (vocab verified). Fails Items 14–16 (single seed), 24 (cost). Overall: **24/30 pass**.

Acknowledgments

Computing resources provided by the NSTDA Supercomputer Center (ThaiSC) on the Lanta HPC system (8.15 PFlops, HPE Cray EX).

Data Availability

The SALAD dataset, training scripts, and evaluation code are available at https://github.com/nutthakorn7/soc-llm-research. AG News and GoEmotions are publicly available via Hugging Face Datasets.

Ethical Statement

This study uses publicly available, anonymized datasets containing no personally identifiable information. No human subjects were involved. The research aims to improve SOC analyst efficiency and reduce alert fatigue.

Declaration of Competing Interest

The author declares no conflict of interest.

References

[1] E. Aghaei et al., "SecureBERT: A Domain-Specific Language Model for Cybersecurity," in *Proc. SecureComm*, 2023.
[2] D. Arp et al., "Dos and Don'ts of Machine Learning in Computer Security," in *Proc. USENIX Security*, 2022.
[3] C. Catal et al., "A Survey of Network Intrusion Detection Datasets," *Computers & Security*, vol. 119, 2022.
[4] Z. Song et al., "CS-Eval: A Comprehensive Large Language Model Benchmark for Cybersecurity," *arXiv:2411.16300*, 2024.
[5] M. Rahman et al., "CTIBench: A Benchmark for Evaluating LLMs in Cyber Threat Intelligence," in *Proc. RAID*, 2024.
[6] M. Bhatt et al., "CyberSecEval 2: A Wide-Ranging Cybersecurity Evaluation Suite for LLMs," *arXiv:2404.13161*, 2024.
[7] T. Dettmers et al., "QLoRA: Efficient Finetuning of Quantized Language Models," in *Proc. NeurIPS*, 2023.
[8] J. Devlin et al., "BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding," in *Proc. NAACL*, 2019.
[9] K. Ethayarajh et al., "Utility is in the Eye of the User: A Critique of NLP Leaderboards," in *Proc. NeurIPS*, 2022.
[10] M. A. Ferrag et al., "Generative AI and Large Language Models for Cyber Security: All Insights You Need," *IEEE COMST*, 2024.
[11] Z. Gekhman et al., "Does Fine-Tuning LLMs on New Knowledge Encourage Hallucinations?" in *Proc. EMNLP*, 2024.
[12] T. K. Ho and M. Basu, "Complexity Measures of Supervised Classification Problems," *IEEE TPAMI*, vol. 24, no. 3, pp. 289–300, 2002.
[13] E. Hu et al., "LoRA: Low-Rank Adaptation of Large Language Models," in *Proc. ICLR*, 2022.
[14] S. Kapoor and A. Narayanan, "Leakage and the Reproducibility Crisis in ML-Based Science," *Patterns*, vol. 4, no. 9, 2023.
[15] D. Kiela et al., "Dynabench: Rethinking Benchmarking in NLP," in *Proc. NAACL*, 2021.
[16] P. Liang et al., "Holistic Evaluation of Language Models," *arXiv:2211.09110*, 2022.
[17] Y. Liu et al., "SecBench: A Multi-Dimensional Benchmarking Dataset for LLMs in Cybersecurity," *arXiv:2412.20787*, 2024.
[18] F. N. Motlagh et al., "Large Language Models in Cybersecurity: State-of-the-Art," *arXiv*, 2024.
[19] N. Moustafa and J. Slay, "UNSW-NB15: A Large-Scale Data Set for NIDS," in *Proc. MilCIS*, 2015.
[20] M. Ring et al., "A Survey of Network-based Intrusion Detection Data Sets," *Computers & Security*, vol. 86, pp. 147–167, 2019.
[21] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset," in *Proc. ICISSP*, 2018.
[22] S. Swayamdipta et al., "Dataset Cartography: Mapping and Diagnosing Datasets with Training Dynamics," in *Proc. EMNLP*, 2020.
[23] N. Tihanyi et al., "CyberMetric: A Benchmark Dataset for Evaluating LLMs' Knowledge in Cybersecurity," *arXiv:2402.07688*, 2024.
[24] A. Vaswani et al., "Attention Is All You Need," in *Proc. NeurIPS*, 2017.
[25] A. Wang et al., "GLUE: A Multi-Task Benchmark and Analysis Platform for NLU," in *Proc. ICLR*, 2019.
[26] A. Wang et al., "SuperGLUE: A Stickier Benchmark for General-Purpose Language Understanding Systems," in *Proc. NeurIPS*, 2019.
[27] N. Chalaemwongwan, "Mind the Label Gap: How Fine-Tuned LLMs Hallucinate Sub-Categories in SOC Alert Classification," *IEEE TDSC*, 2025.
[28] N. Chalaemwongwan, "1,000 Labels Is All You Need: Non-Monotonic Scaling of Label Compliance," *ESWA*, 2025.
[29] N. Chalaemwongwan, "Higher Rank, More Hallucination? LoRA Capacity and Label Compliance," *Neurocomputing*, 2025.